E-Scooter Speed Safety in South Korea

Project Progress: From Raw GPS to Submission-Ready Manuscript

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Presentation Outline

Part I — Data & Exploration

1. Project Motivation
2. Data Preprocessing (Phase 1)
3. Speed Indicators (Phase 2)
4. Map-Matching & Road Class (Phase 3)
5. Spatial Analysis (Phase 4)

Part II — Modeling & Paper

6. Statistical Modeling (Phase 5)
7. Extension Analyses (Phase 8)
8. DiD Causal Analysis (Phase 11)
9. Survival Analysis (Phase 12)
10. Paper Status & Next Steps

Target venue: *Transportation Research Part C*

101/107 tasks complete

Project Motivation

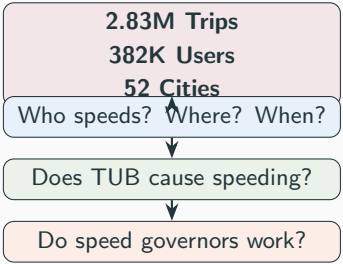
Why E-Scooter Speed Safety?

Context

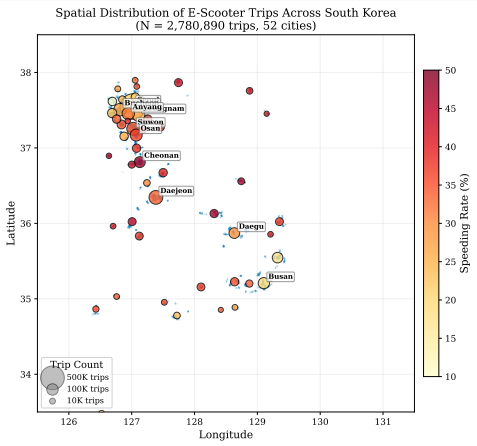
- E-scooters: fastest-growing micro-mobility mode
- South Korea: 25 km/h speed limit
- Swing: largest Korean operator
- Modes: **TUB** (turbo), **STD**, **ECO**

Research Gaps

- No large-scale GPS study for e-scooter speeding
- Mode-specific speed governance poorly understood
- Causal evidence on speed limiters missing



Dataset Overview



Swing GPS Trajectories

- **Cross-section**: 2.83M trips (May 2023)
- **Longitudinal**: 44.1M trips (24 months)
- Speed data: Feb–Dec 2023 only
- 25 columns: GPS, speeds, mode, model

Key characteristics

- Sensor-based speeds (not GPS-derived)
- GPS interval $\sim$10s; sentinel -999 for NULL
- 17 provinces, 52 cities

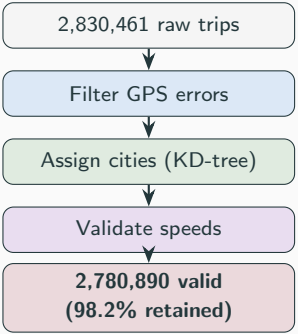
Phase 1: Data Preprocessing

Phase 1: Data Quality Control

Why: Raw GPS data contains errors and inconsistencies that would bias all downstream analyses.

Pipeline (7 tasks)

1. Profile schema: null rates, types, ranges
2. Filter: 19 outliers (>50 km/h), zero-distance, <5 GPS points
3. City assignment: KD-tree to 54 city centers
4. Per-city summary statistics
5. Validate speeds vs. displacement-derived
6. Build cleaned Parquet dataset
7. Data quality report



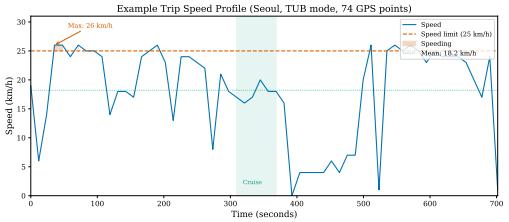
Phase 2: Speed Indicators

Phase 2: Computing Speed Safety Indicators

Why: Transform raw speed arrays into interpretable metrics for modeling.

Trip-level indicators

- Mean/max/P85 speed, speed CV
- Speeding rate: fraction >25 km/h
- Harsh accel threshold: **0.5 m/s^2** (10s intervals)
- Profile: ramp-up, cruise, zero-speed fractions



User-level aggregates

- Speeding propensity, speed consistency
- Harsh events, temporal/spatial patterns

Phase 3: Road Network Assignment

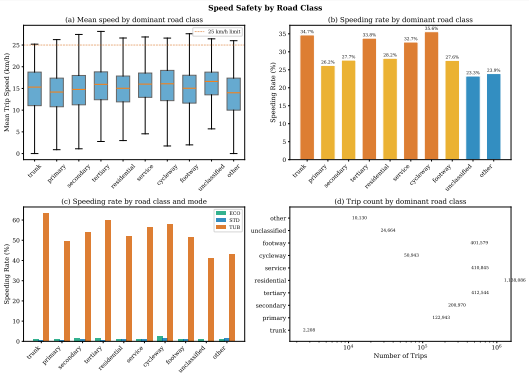
Phase 3: Map-Matching & Road Class Assignment

Why: Link GPS to road infrastructure to study road-type effects on speeding.

Approach evolution

1. OSM networks: 50 cities, 1.32M nodes
2. LeuvenMapMatching: 5.8s/trip (infeasible)
3. mappymatch: API incompatible
4. **DuckDB + KDTree**: 105M pts in 463s

Distribution: residential 41%, tertiary 15%, service 15%, footway 14.5%, major 13.6%



Phase 4: Spatial Analysis

Phase 4: Where Does Speeding Cluster?

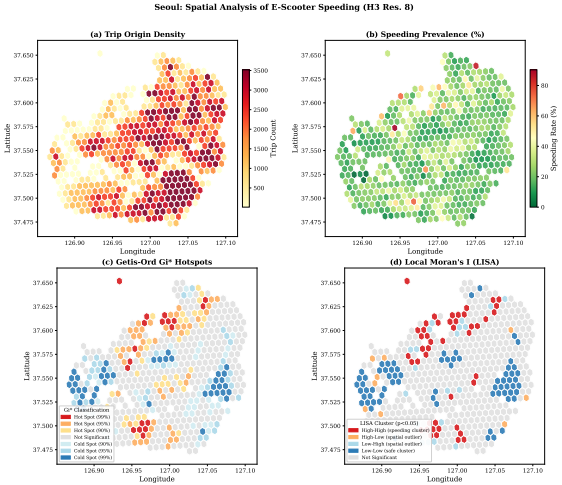
Why: Identify hot spots for targeted enforcement and infrastructure.

Methods

- H3 hex grid (res. 8, ∼461m): 6,111 hexes
- Getis-Ord Gi* ($k = 8$, 999 perms)
- Global Moran's I + LISA clusters

Key findings

- Seoul: 101 hot / 123 cold spots
- Hot spots: lower cycling infra
- Daegu: I = 0.712 (highest)



Phase 5: Statistical Modeling

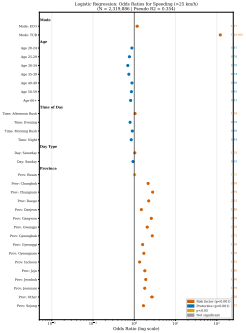
Phase 5: Model Suite Overview

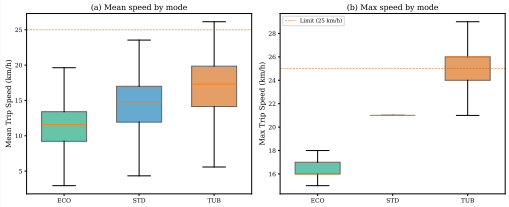
Why: Quantify speeding drivers, accounting for user heterogeneity and zero-inflation.

#	Model	Purpose	Key Finding
1	Logistic + GEE	Speeding probability	TUB OR = 121
2	Mixed-effects LM	Mean speed drivers	ICC = 0.271
3	Two-part hurdle	Zero-inflated speeding	70.4% zeros
4	GMM ($K=4$)	Rider typology	4 clusters
5	Multinomial logit	Class membership	TUB RRR = 11.8
6	GEE experience	Learning effect	OR = 1.18/log-trip
7	Spline logistic	Distance–speeding	pseudo-R^2=0.39
8	SHAP/LightGBM	Feature importance	AUC = 0.905

All models: TUB mode is the dominant predictor of speeding.

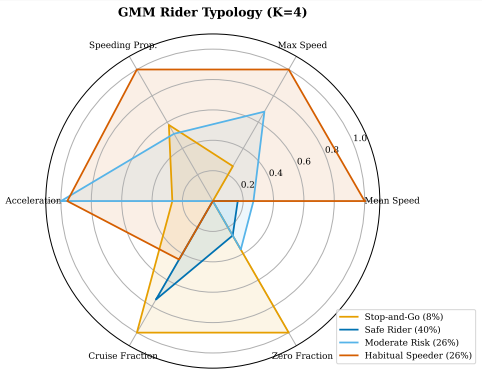
Core Finding: TUB Mode Dominates





- TUB mode OR = 121 for speeding; ECO → 34% speed reduction, speeding 54% → 1.1%
- Within-subject paired ($n = 12{,}046$): all effects $p < 0.001$

Rider Typology: Four Behavioral Clusters

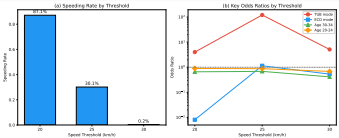


GMM $K = 4$: Safe (40%), Moderate (26%), Habitual Speeder (26%), Stop-and-Go (8%)

Multinomial logit (class membership):

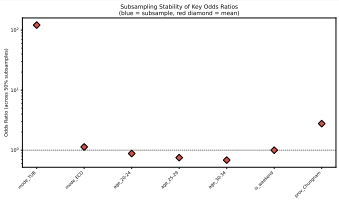
- TUB: RRR = 11.8 for Speeder class
- ECO: RRR = 0.06 (protective)
- Pseudo $R^2 = 0.297$

Robustness: Findings Hold Across Tests



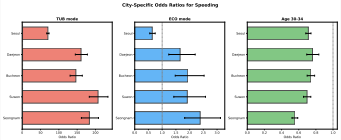
Threshold sensitivity

20 km/h: 87%
25 km/h: 30%
30 km/h: 0.2%



Subsampling stability

TUB OR CV $< 1.2\%$
across 5 folds



City-specific models

TUB OR range: 70–207
across 5 cities

Phase 8: Extension Analyses

Phase 8: Five Extension Research Questions

Why: Extend to 24-month longitudinal data (44.1M trips) for deeper insights.

8.1 Experience effect

- Speeding: 3.8% (trip 1) $\rightarrow$ 28% (500+)
- But: via TUB adoption, not skill
- TUB use: 11.5% $\rightarrow$ 47.8% over 50 trips

8.2 Trip distance

- 13.9% (149m) $\rightarrow$ 54.2% (2168m)
- Speed drift: -1.5 km/h in 2nd half

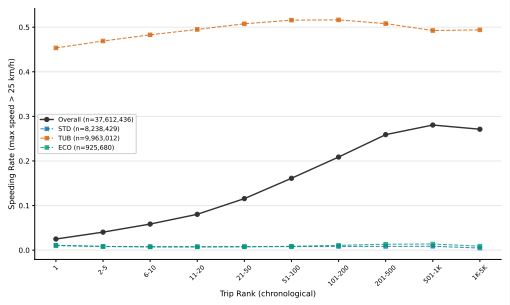
8.3 Road curvature

- More curves $\rightarrow$ less speeding (OR=0.716)
- Risk 5$\times$ higher when speeding on curves

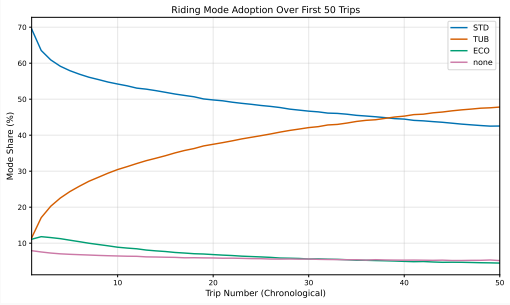
8.4 SHAP interpretability

- LightGBM AUC = 0.905
- TUB SHAP = 2.33 (3.6$\times$ next)

Experience Effect: It's About Mode Adoption



Speeding rate by experience level

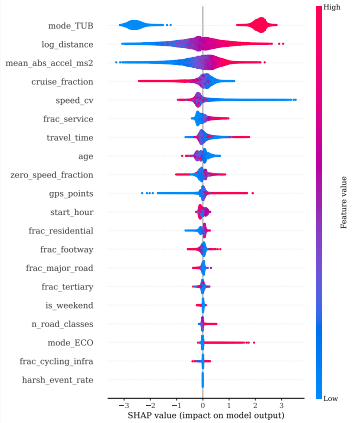


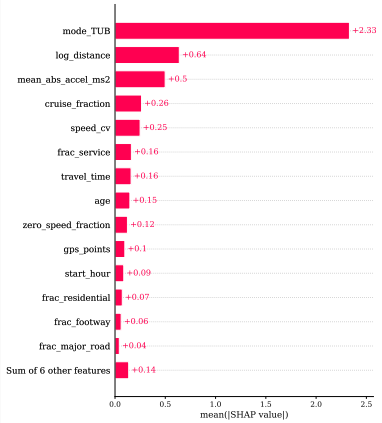
TUB adoption over first 50 trips

Within STD/ECO modes only: speeding is virtually constant across experience (0.8–1.5%).

$\Rightarrow$ **98.4% of the experience–speeding gradient is mediated by TUB adoption.**

SHAP: TUB Mode Dwarfs All Other Features



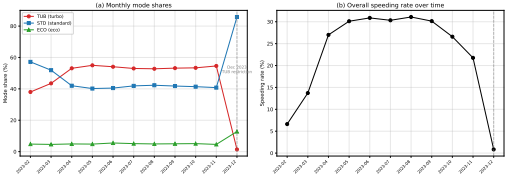


TUB mean |SHAP| = 2.33 vs. 0.64 (log distance) — 3.6× gap.

Phase 11: DiD Causal Analysis

Phase 11: The December 2023 Natural Experiment

Why: From association to **causation**. Dec 2023 TUB ban = clean quasi-experiment.



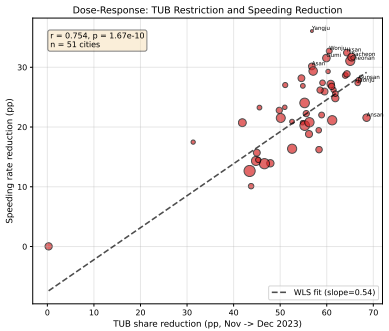
Treatment: System-wide TUB ban

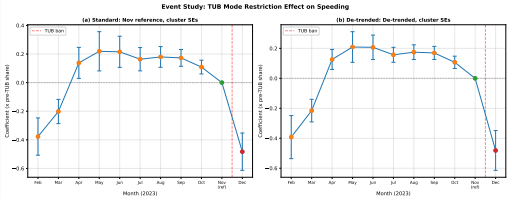
- TUB: 54.6% → 1.4%
- STD absorbed: 40.8% → 85.8%
- 50/54 cities >10pp TUB drop

ID: Cross-city TUB share variation

Panel: 53 cities × 11 months

DiD Results: Causal Speed Reduction

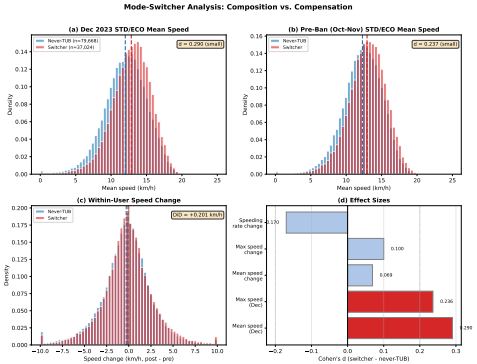




- **TWFE**: $\beta = -0.568$ — 50% pre-TUB city $\rightarrow$ 28.4pp reduction
- **Dose-response**: 10pp TUB $\rightarrow$ 5.0pp speeding ($r = 0.754$)
- **Placebo**: $\beta = -0.015$, $p = 0.561$ — **PASS**

Behavioral Substitution: No Risk Compensation

Why: Does banning TUB make riders speed more in STD? (Risk compensation)



Multi-outcome DiD

- Overall speeding: $\beta = -0.568$ ✓
- Trip distance: n.s. ($p = 0.45$)
- STD/ECO speeding: n.s. ($p = 0.59$)
- STD/ECO speed: +0.85 km/h (composition)

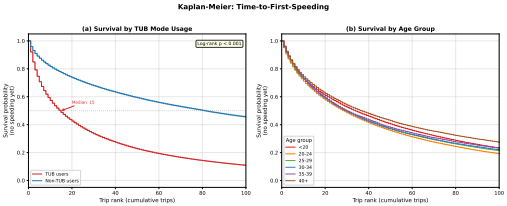
Mode-switcher ($n = 120$K)

- Within-user: +0.20 km/h ($d = 0.069$)
- **Verdict**: Composition, not compensation

Phase 12: Survival Analysis

Phase 12: Time-to-First-Speeding

Why: When do riders first cross 25 km/h? Survival analysis reveals onset dynamics.



864K users ($\geq$2 trips): 48% event, 52% censored

- TUB: median **15 trips** to first speeding
- Non-TUB: median **81 trips** (5.4$\times$)
- Log-rank $\chi^2 = 91{,}282$
- Cox TV: TUB HR = **10.41**

98.4% of experience–speeding gradient mediated by TUB adoption

Paper Status & Next Steps

Paper: Submission-Ready Manuscript

Status

- ~8,945 words (TR-C: 8–12K)
- 79 pages, 0 LaTeX errors
- 9 models + DiD + survival
- 49+ figures, 51 bib entries (100% verified)
- Overleaf synced via git subtree

QA: 49% of bib entries corrected; 3 unverifiable refs replaced; orphans removed

All sections complete

✓ Abstract & Highlights

✓ Introduction (8 contributions)

✓ Literature Review

✓ Data Description (2 tables)

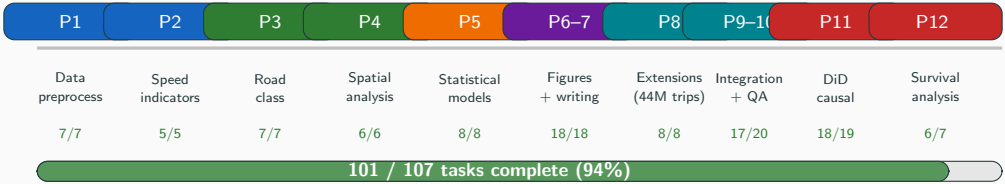
✓ Methodology (9 models)

✓ Results (8 subsections)

✓ Discussion (5 policy recs)

✓ Conclusions + Appendix

Progress Timeline



Remaining Items & Next Steps

Remaining (6 tasks)

- Co-author confirmation (PI)
- Cover letter (PI)
- Upload to Elsevier (PI)
- Include DiD? (PI)
- Integrate survival into text
- Final pre-submission

Key contributions

1. Largest GPS e-scooter study (2.83M)
2. TUB OR = 121 for speeding
3. Causal: TUB ban $\rightarrow$ 28.4pp reduction
4. No risk compensation
5. 98.4% experience via TUB adoption
6. Curvature risk paradox
7. 4 rider typologies
8. Speed governors work

Thank you

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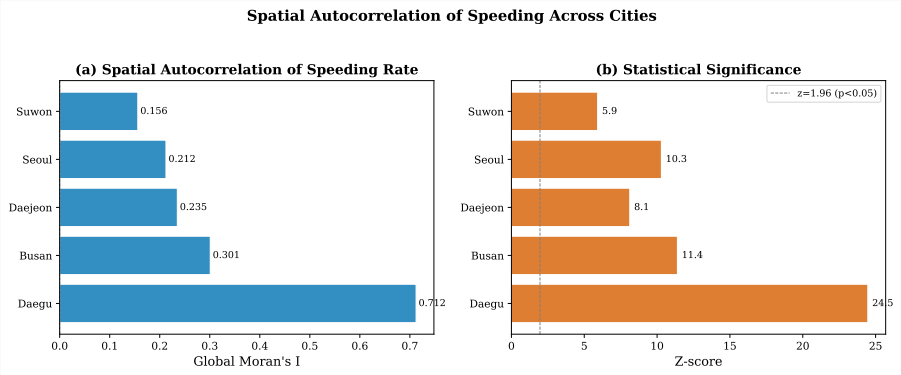
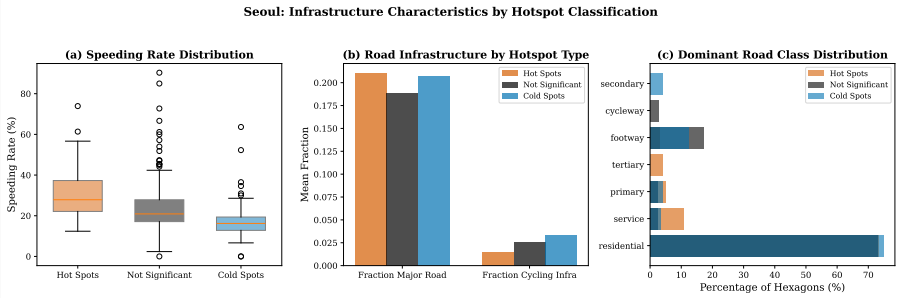
Target: Transportation Research Part C

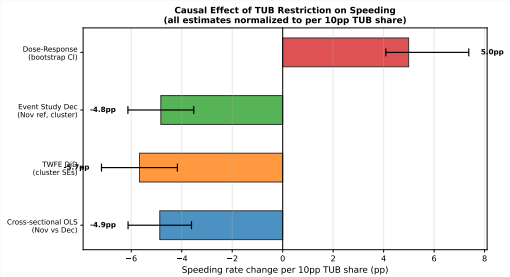
Backup: Moran's I Comparison Across Cities



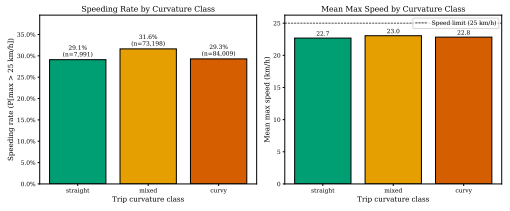
Backup: Infrastructure Overlay

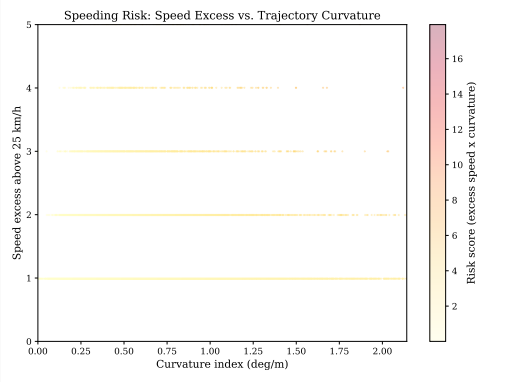
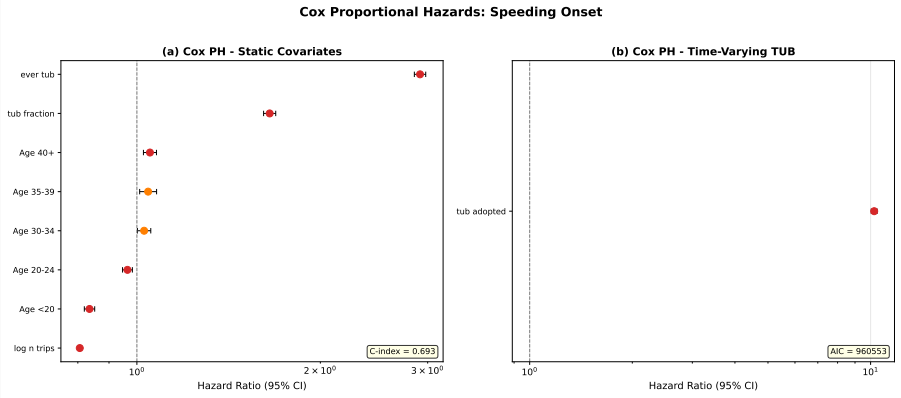


Backup: DiD Robustness



Backup: Curvature Risk Paradox





Backup: Cox Proportional Hazards